



NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech

Internal Examinations, February-2026

Course Name: Health & Happiness

Date: 17-02-2026

Duration: 30 Minutes

Course Code: HU151

Time: 11:30 – 12:00

Max. Marks: 15

Roll No

2

5

E

C

E

1

4

3

7

ANSWER ALL QUESTIONS

A. MULTIPLE CHOICE QUESTIONS ($0.5 \times 10 = 5$)

1. Which function of food is MOST closely linked to social behavior?
 - A. Energy production
 - B. Immunity building
 - ☒ C. Psychological satisfaction
 - D. Tissue repair
2. According to WHO, health excludes which of the following?
 - A. Mental well-being
 - B. Social well-being
 - ☒ C. Emotional happiness
 - D. Absence of disease alone
3. Which lipid is a major component of cell membranes?
 - A. Triglycerides
 - B. Waxes
 - ☒ C. Phospholipids
 - D. Steroids
4. The structural unit of fats stored in the body is:
 - A. Fatty acid
 - B. Phospholipid
 - ☒ C. Triglyceride
 - D. Glycolipid
5. Which vitamin deficiency may lead to night blindness?
 - A. Vitamin D
 - ☒ B. Vitamin A
 - C. Vitamin K
 - D. Vitamin E
6. Which phytochemical group helps reduce blood pressure?
 - A. Carotenoids
 - B. Polyphenols

- C. Phytosterols
- D. Isoflavones
- 7. Isoflavones are mainly found in:
 - A. Fruits
 - B. Cereals
 - ☒ C. Legumes
 - D. Spices
- 8. Which spice shows strong anti-microbial AND anti-fungal activity?
 - A. Cinnamon
 - B. Ginger
 - ☒ C. Coriander
 - D. Cardamom
- 9. Happiness is considered "naturally acceptable" when it involves:
 - A. Competition
 - B. Opposition
 - C. Good relationships
 - D. Isolation
- 10. Which phytochemical gives red and purple colour to berries?
 - A. Flavonoids
 - B. Anthocyanidins
 - C. Carotenoids
 - D. Polyphenols

B. SHORT ANSWER QUESTIONS (2*5=10)

1. Why are macronutrients required in larger quantities than micronutrients?
2. Mention any two long-term benefits of good nutrition.
3. Differentiate between *lipids* and *fats* (any two points).
4. Define phytochemicals and state one key function.
5. Importance of Nasya.



National Institute of Technology Goa

Kottamoll Plateau, Cuncolim, Salcete, South Goa, Goa - 403703

Programme Name: BTech

Mid Semester, Mar. 2026

Roll No 2 5 E C E 1 0 3 7

Course Name : Health & Happiness

Course Code : HU151

Date : March 5, 2026

Time Duration : 10:00 a.m. - 11:30 a.m.

Duration : 1 Hr. 30 Min.

Max. Marks : 50

- Answer all the questions neat & cleanly. **One (1) mark is considered for neat and clean Answer script.**
- No marks is there for irrelevant text/answer.
- **Any type of malpractice invites the cancellation of your answer script.**

Part - A

1. Choose the most appropriate answer from the options provided below.

[1 × 15 = 15]

- (a) The basic structural and functional unit of the human body is:
- Tissue
 - Organ
 - Cell
 - System
- (b) Which nutrient is the primary source of energy for the body?
- Proteins
 - Vitamins
 - Carbohydrates
 - Minerals
- (c) Proteins are made up of:
- Fatty acids
 - Amino acids
 - Glucose
 - Minerals
- (d) Unsaturated fats are generally:
- Solid at room temperature
 - Liquid at room temperature
 - Harmful to health
 - Absent in plant sources
- (e) Which vitamin is known as the "sunshine vitamin"?
- Vitamin A
 - Vitamin C
 - Vitamin D
 - Vitamin K
- (f) Iodine deficiency causes:
- Night blindness
 - Diabetes
 - Goiter
 - Hypertension
- (g) Balanced diet means:
- Equal portions of junk and healthy food
 - Eating any food in large quantities
 - Including all essential nutrients in proper amounts
 - Avoiding fats completely
- (h) Lipids are defined as:
- Organic substances insoluble in water but soluble in organic solvents
 - Simple carbohydrates
 - Inorganic substances soluble in water
 - Proteins with nitrogen
- (i) Which of the following practices helps maintain a healthy circadian rhythm?
- Sleeping at irregular times
 - Late-night screen exposure
 - Fixed sleep and wake-up schedule
 - Skipping breakfast daily

- (j) Kapha dosha is mainly related to:
- i. Heat
 - ii. Movement
 - iii. Structure and stability
 - iv. Vision
- (k) Samadhi represents:
- i. Physical exercise
 - ii. Monounsaturated fat
 - iii. Hydrogenated fat
 - iv. None of these
- (l) The term circadian rhythm refers to:
- i. The process of digestion quantities
 - ii. The heart's pumping cycle
 - iii. A 24-hour biological clock controlling body functions
 - iv. The rhythm of breathing
- (m) Niyama refers to:
- i. Social discipline
 - ii. Moral restraints
 - iii. Personal observances
 - iv. Physical postures
- (n) Which limb of Ashtanga Yoga involves concentration on a single object?
- i. Pratyahara
 - ii. Dharana
 - iii. Dhyana
 - iv. Samadhi
- (o) Which of the following is a component of a healthy lifestyle?
- i. Sedentary habits
 - ii. Balanced nutrition
 - iii. Excessive junk food
 - iv. Chronic stress

Part - B

2. Answer the following question. [2 × 5 = 10]
- (a) Differentiate between saturated and unsaturated fats (any two points).
 - (b) Write any four health benefits of spices commonly used in Indian cooking.
 - (c) What is the meaning of Gandusha?
 - (d) What are the basic food groups? Name any four food groups included in a balanced diet.
 - (e) How does the practice of Ashtanga Yoga contribute to physical and mental health?

Part - C

3. Write a Short Note on each of the following topics. [3 × 5 = 15]
- (a) Yama and Niyama
 - (b) Circadian Rhythm
 - (c) Importance of Yoga in Modern Life
 - (d) Dinacharya
 - (e) Ustrasana (Yoga) & its Importance.

Part - D

4. Define Prakriti. Name the types of Doshas in Ayurveda. Explain the characteristics of *kapha prakriti* and *pitta prakriti*. [2 + 2 + 5 = 9]

----- All The Best -----



Roll No 2 5 E C E 1 0 3 7

NATIONAL INSTITUTE OF TECHNOLOGY GOA

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.

End Semester Examinations, May-2026

Course Name: Health & Happiness

Date: May 4, 2026

Duration: 3 Hours

Course Code: HU151

Time: 10:00a.m. - 1:00p.m.

Max. Marks: 100

ANSWER ALL QUESTIONS

- One (1) mark is considered for neat and clean Answer script.
- No marks are there for irrelevant text/answer.
- Any type of malpractice invites the cancellation of your answer script.

A. Multiple Choice Questions [25*1=25]

1. Proper handwashing helps prevent:
 - a) Ripening
 - b) Food contamination
 - c) Fermentation
 - d) Packaging
2. GM foods are produced by:
 - a) Natural selection
 - b) Traditional farming only
 - c) Genetic engineering
 - d) Fermentation
3. Which crop is modified for high Vitamin A content?
 - a) Wheat
 - b) Rice
 - c) Barley
 - d) Oats
4. HACCP stands for:
 - a) Hazard Analysis and Critical Control Points
 - b) Health and Care Control Process
 - c) Hazardous Chemical Control Plan
 - d) Healthy Agricultural Crop Control Program
5. Which of the following is a risk of GM food?
 - a) Increased nutrition only
 - b) Antibiotic resistance
 - c) Better shelf life only
 - d) Faster cooking
6. Proteins are made up of:
 - a) Fatty acids
 - b) Amino acids
 - c) Glucose
 - d) Minerals
7. The primary structure of protein refers to:
 - a) Folding pattern

- b) Sequence of amino acids
 - c) 3D shape
 - d) Multiple chains
8. Hemoglobin is a:
- a) Fibrous protein
 - b) Globular protein
 - c) Vitamin
 - d) Mineral
9. Denaturation of proteins can occur due to:
- a) Cooling only
 - b) Heat and pH changes
 - c) Sunlight only
 - d) Sugar addition
10. Which vitamin is fat-soluble?
- a) Vitamin C
 - b) Vitamin B1
 - c) Vitamin D
 - d) Vitamin B12
11. Vitamin K is important for:
- a) Vision
 - b) Blood clotting
 - c) Digestion
 - d) Hearing
12. According to the framework of internal state, Chitta is composed of:
- a) Blood, skin, and bones
 - b) Mind, intellect, memory, and ego
 - c) Food, air, and water
 - d) Past, present, and future
13. Expiry date mainly relates to:
- a) Decoration
 - b) Product safety
 - c) Marketing
 - d) Transportation
14. Which pose develops concentration and balance in Moon Salutation?
- a) Dandasana
 - b) Bhujangasana
 - c) Ardha Chandrasana
 - d) Padahasthasana
15. Which cooking method involves cooking food in an enclosed space where heat is evenly distributed without using water?
- a) Boiling
 - b) Baking
 - c) Grilling
 - d) Poaching
16. In the broiling technique, the heat is applied to the food from:
- a) The bottom
 - b) The sides
 - c) Above the food
 - d) A vacuum seal

17. The Food Safety and Standards Act in India was established in which year?
- 1955
 - 1992
 - 2006
 - 2011
18. Which method involves cooking food quickly at a high temperature so that a caramelized crust forms?
- Baking
 - Searing
 - Poaching
 - Steaming
19. Which practice is used in Hatha yoga to achieve a state of deep relaxation and mindfulness?
- Sun Salutation
 - Yoga Nidra
 - Cobra Pose
 - Fasting
20. Which of the following is considered a sign of good health?
- Difficulty awakening in the morning
 - Frequent abnormal dreams
 - Relish at mealtime and proper digestion
 - A very high resting pulse rate
21. Vitamin C helps in:
- Collagen synthesis
 - Blood pressure only
 - Hearing
 - Hair coloring
22. A feeling that is naturally acceptable and leads to happiness in a relationship is:
- Mistrust
 - Jealousy
 - Trust
 - Hate
23. Human biology is an interdisciplinary study that examines humans through:
- Physics and chemistry only
 - Genetics, evolution, physiology, and nutrition
 - History and literature only
 - Accounting and finance
24. What is the generally accepted normal blood pressure for an adult?
- 140/90 mmHg
 - 100/60 mmHg
 - 120/80 mmHg
 - 150/95 mmHg
25. The normal body temperature range for a healthy adult is:
- 95.0 to 96.5 degrees Fahrenheit
 - 97.8 to 99.1 degrees Fahrenheit
 - 100.2 to 101.5 degrees Fahrenheit
 - 102.0 to 103.5 degrees Fahrenheit

B. Short Answer Questions [10*2=20]

1. Describe the "Sous Vide" technique and its benefits.
2. How does Hatha yoga improve mental focus and clarity?
3. List two specific food laws or orders in India besides the FSSAI Act 2006.
4. What are "Vital Signs," and why are they measured?
5. Define post-harvest loss
6. What is the main aim of HACCP?
7. What are nutrients?
8. Define primary structure of protein.
9. What is Broiling?
10. Define systolic and diastolic blood pressure.

C. Write a Short Note on each of the following topics [6*4=24]

1. Sadvritta
2. Phytochemicals and its Health Benefits
3. Natural Urges
4. Genetically Modified Foods and Their Associated Risks
5. Distinguish between Fat Soluble Vitamins and Water Soluble Vitamins
6. Fast Food and its effect on your body

D. Long Answer Questions [(2+2+4)*3=24]

1. What is Rutujacharya? Briefly explain the Sharad Ritu Diet. Explain the Sharad Ritu Lifestyle.
2. What is Ashtanga Yoga and Hatha Yoga? Difference between Ashtanga Yoga and Hatha Yoga? What are the Key components of Ashtanga Yoga?
3. What is Food Adulteration and Food Contamination? What are the types of Food Contamination? How to prevent Food Contamination?

E. Descriptive Question [1*6=6]

1. Describe the procedure of Dhanurasana & Briefly discuss about its advantages